Multiphysical Interpretable Deep Learning Network for Oil Spill Identification Based on SAR Images

Jianchao Fan, *Member, IEEE*, Zitai Sui, and Xinzhe Wang

***Abstract*— The application of deep learning algorithms to oil spill identification in synthetic aperture radar (SAR) remote sensing images has enabled substantial progress. However, the end-to-end learning approach of deep learning has inefficient interpretability, making it difficult to ensure the reliability of oil spill identification tasks. In addition, how to effectively apply the physical information of SAR images to make intelligent decoding methods more interpretable is the key to understanding and evaluating its oil spill identification results. To address these issues, this article proposes a multiphysical interpretable deep learning network (MIDLN) for oil spill identification based on SAR images. MIDLN defines a dual neural network consisting of a multiphysical deep convolutional neural network and an adaptive grad-weight selection class activation explainer. The explainer adaptively selects the most important interpretation weights for the category, mitigating visual background noise in the explainer results. The high-quality results of the explainer are utilized to provide feedback signals to the network, and the learning strategy of the network is adjusted to structure a deep learning framework with interpretable feedback. Meanwhile, to enhance the physical interpretability of the network, this study employs a scribble-style interactive sampling to streamline the oil spill physical information extraction process. The effectiveness of two novel oil spill physical features is validated through the combination of pixel-level oil–water feature analysis and deep learning methods. In addition, a multiphysical feature extraction head is designed for different features to enhance the uniqueness of physical information and reduce the redundant interference of multiple features. The superiority of the proposed method over existing algorithms is demonstrated by experiments on a large number of Sentinel-1 oil spill datasets. The code for this work will be made available at https://github.com/fjc1575/Marine-Oil-Spill/tree/main/MIDLN for the sake of reproducibility.**

***Index Terms*— Interpretable deep learning, interpretable feedback, multiphysical feature extraction, oil spill identification, synthetic aperture radar (SAR).**

I. Introduction

MARINE oil spills can result from ship accidents, ruptures of submarine oil pipelines, explosions of oil rig platforms, and deliberate discharge of tank-cleaning wastewater from ships. Oil spill occurrence varies in scale and often causes severe seawater pollution, with catastrophic consequences for marine ecosystems [1], [2]. According to statistics, more than two million tons of oil are spilled annually into the oceans [3]. Therefore, the timely development of the oil spill damage assessment and diffusion control is essential for marine environmental protection [4], [5]. Synthetic aperture radar (SAR) has the advantage of being all-day and unaffected by cloud cover and other conditions [6], [7], [8]. It has gradually become a favorable tool for oil spill observation. The oil slick reduces the backscattering coefficient of the sea surface [9], resulting in it being a dark spot in SAR images [10]. However, oil spill areas typically occupy only a tiny portion of SAR images, and their shapes are highly irregular due to the influence of seawater. Therefore, developing effective oil spill detection techniques is crucial for accurately recognizing oil spills in SAR images.

Researchers have continuously developed feature extraction methods, considering the importance of feature extraction in SAR oil spill detection. The relevant features should demonstrate a clear relationship between the selected features' values and an oil spill's probability. These features can be categorized into geometric, textural, polarimetric, and physical features. Physical features describe the backscattering values of oil spills and their surrounding environment and are considered the key for distinguishing oil spills from alook-likes [11], [12]. Mdakane and Kleynhans [13] extracted 29 oil spill features. They input these features into the network to evaluate their influence on the results while ranking the importance of features. Chen et al. [14] obtained various oil spill features by considering the internal distribution of the backscattering value in SAR images, improving the segmentation results of oil spills. However, the feature above extraction methods still have the problem of cumbersome extraction processes. Meanwhile, oil spills exhibit a high degree of heterogeneity [15], [16]. In the face of oil spill identification with multifeature fusion, traditional networks may not adequately consider the distinctiveness of different features, thereby negatively impacting identification results.

Deep learning has succeeded in different remote sensing imagery applications [17], [18]. It possesses robust learning representation capabilities, enabling the automatic mining of information from vast amounts of data [19]. Convolutional neural networks (CNNs) [20] are the classic structure in deep learning networks [21]. There are generally two types of object recognition algorithms based on CNNs: two-stage object detectors and one-stage object detectors. Among the

Manuscript received 5 November 2023; revised 3 January 2024; accepted 19 January 2024. Date of publication 24 January 2024; date of current version 7 February 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42076184, Grant 41876109, and Grant 41706195; in part by the National Key Research and Development Program of China under Grant 2021YFC2801000; in part by the National High Resolution Special Research under Grant 41-Y30F07-9001-20/22; in part by the Fundamental Research Funds for the Central Universities under Grant DUT23RC(3)050; and in part by the Dalian High Level Talent Innovation Support Plan under Grant 2021RD04. *(Corresponding author: Xinzhe Wang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3357800

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two-stage object detectors, the R-CNN series, which includes Faster R-CNN [20], is the most representative. Huang et al. [22] identified oil spill areas in large-scale SAR remote sensing images by analyzing additional factors such as environment and combining them with the Faster R-CNN object detection framework. Two-stage object detectors typically offer higher accuracy but are slower, making them less suitable for oil spill detection. The most representative of one-stage target detectors is the You Only Look Once (YOLO) version [23] series. YOLO is typically much faster than two-stage object target detectors but tends to have lower detection performance. The subsequent variants of the YOLO series, such as YOLOv4 [24], YOLOv5 [25], and YOLOx [26], enhance detection accuracy by leveraging robust backbone networks. Yang et al. [27] employed the YOLOv4 model to learn oil spill features within the model's region of interest and the background information of SAR images and effectively validated it on extended datasets. Fan et al. [28] designed a YOLOX-based oil spill detection network using various SAR oil spill physical features. The network significantly improves the identification accuracy of long strip-shaped oil spill areas through the complementary use of multiple physical features. YOLOv7 [29] is an efficient target recognition algorithm with optimized modules and training methods. In complex scenarios of SAR oil spill images, YOLOv7 delivers fast speed and high accuracy, making it more suitable for oil spill detection. Nevertheless, traditional deep learning end-to-end learning modes face problems such as inefficient interpretability, which makes it difficult to ensure reliability in oil spill identification tasks.

Besides the identification capability, the interpretability of the model is another crucial aspect of neural networks [30]. Introducing a class activation map (CAM) [31] in deep learning is a common method to enhance the interpretability of algorithms. This approach assigns appropriate contribution weights to features in the network and generates a heatmap through linear summation to explain the network's decisions. Grad-CAM [32], as an iterative enhancement of CAM, was applied to a wide range of CNN models and was no longer constrained by a fully connected layer requirement. However, its interpretation is incomplete when dealing with multiple targets. To address this limitation, Grad-CAM++ [33] improved the interpretation of various objects of the same class in a single image. Compared to Grad-CAM and Grad-CAM++, XGrad-CAM [34] further enhances the visualization effects of interpretable class activation graph networks. The CAM can depict crucial decision regions in networks. However, due to the correlation between the background seawater and oil spill targets and the possible introduction of noise in the network during gradient propagation [35], the result of the class activation is visually noisy, as shown in Fig 1. In contrast to CAMs, attention mechanisms can indirectly enhance the interpretability of models [36]. Attention mechanisms introduce attention weights and learn these weights to determine the importance of each feature for model learning, thereby achieving an internal self-interpretation effect within the model. Given its effectiveness in optical images, many researchers have also applied it to SAR images [37], [38], [39]. Nevertheless, both of these interpretable methods have limitations. While attention mechanisms improve the detection accuracy of the network, they provide no explanation for researchers. CAMs give the network's important decision areas and the features' importance to the outcome; explainer results are limited to post-analysis and cannot be utilized for model feedback to enhance the network performance.

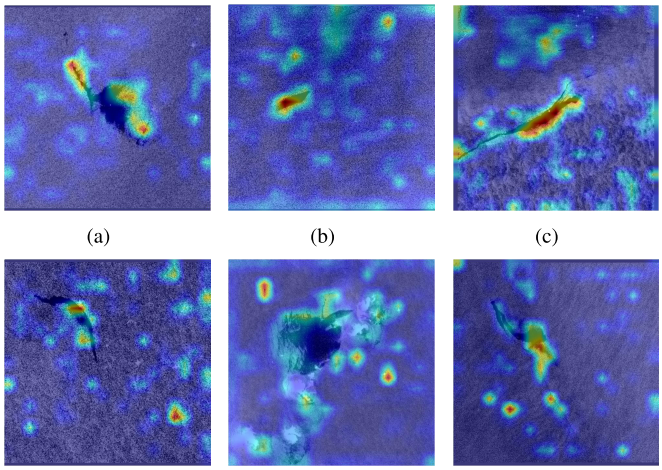


Fig. 1. Due to the correlation between the oil spill morphology and the background seawater (seawater flow leads to irregularities in the oil spill morphology), the network has a lot of gradient noise in the backpropagation. This leads to low-quality explainer results and much visual noise in the class activation maps. (a) and (b) Oil spill class activation explainer results on a bright background. (c) and (d) Class activation explainer results for long strip-shaped oil spill targets. (e) Oil spill class activation explainer results on a complex background. (f) Oil spill class activation explainer results on a dark background.

Targeting the above problems, this article proposes a multiphysical interpretable deep learning network (MIDLN) for interpretable oil spill identification based on SAR images. MIDLN has a dual network, including a multiphysical deep convolutional neural network (MDCNNet) for detecting oil spills and an adaptive grad-weight selection class activation explainer (AGS-CAExplainer) for interpreting the network and improving its performance. First, oil spill saliency information is independently extracted from multiple backscatter-based physical features using a multiphysical feature extraction head (MFEHead) to ensure the uniqueness of each feature. Subsequently, the backbone is adopted for further downsampling. The feature extraction results are fused and utilized for both the idetect head and the AGS-CAExplainer to achieve oil spill detection and explainer output simultaneously. The interpretable decision coefficients calculated by the explainer results are utilized as feedback signals and the sparse window attention model (SWAM) is adjusted to switch the learning strategy of the network. The network is iteratively updated to improve the oil spill identification results.

1) An oil spill identification framework based on interpretable feedback is proposed. It can fully leverage the explainer's results to provide feedback to the network, adjusting the network's learning strategy by calculating the interpretability decision coefficient as a feedback signal. In the early stages of training, important decision areas are focused near oil spill targets by learning oil spill global feature semantic information. In the later stages of training, it supplements the learning of oil spill edge shape detail information for oil spill targets.

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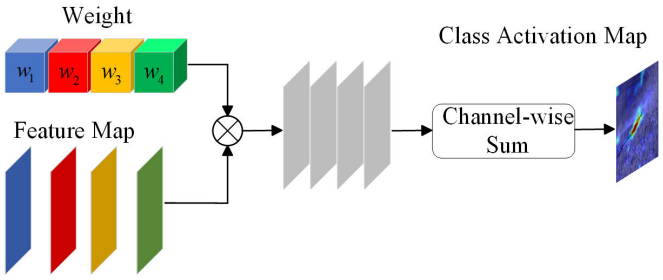


Fig. 2. Process of CAM methods [31].

2) An AGS-CAExplainer is built to improve visual background noise in class activation results. The explainer adaptively selects the most important interpretable weights for the class, resulting in less visual noise and higher quality while retaining the localization information.
3) Two new oil spill physical features, oil film power ratio and seawater power ratio, are proposed to simplify SAR feature extraction methods. The embedding of physical information significantly enhances the model's physical interpretability and reliability. Based on the above features, the MFEHead is designed to extract multiple oil spill physical information. It avoids the feature redundancy interference caused by multiple features sharing convolution, enhancing the uniqueness of physical information and the spatial consistency of oil spill targets.

The remainder of this article is organized as follows. The related work is briefly reviewed in Section II. Section III introduces the proposed MIDLN in detail for SAR detecting oil spill images. Experimental results are discussed in Section IV. Finally, the conclusions are provided in Section V.

II. Related Works

The intuitiveness and effectiveness of CAMs have stimulated the development of interpretability [40]. Fig. 2 shows the process of CAM methods. The explainer in this article is mainly inspired by two algorithms, CAM [31] and Grad-CAM [32]. These two algorithms are currently widely utilized in explaining deep neural networks.

In CAM [31], authors demonstrate that the global average pooling (GAP) layer allows for visualization of the predicted class scores on a given image, highlighting the important decision region part of the neural networks. In neural networks with GAP, the final classification score y^c for class c can be written as a linear combination of its global average pooled last convolutional layer k feature maps A^k

$$y^c = \sum_k w_k^c \cdot \sum_i \sum_j A_{ij}^k \tag{1}$$

where w_k^c are importance weights for each layer of feature maps of class c, and each pixel (i, j) is computed in the CAM L^c as

$$L_{ij}^c = \sum_k w_k^c \cdot A_{ij}^k \tag{2}$$

where L_{ij}^c represents the degree of important decision contribution of each pixel in feature maps A_{ij}^k with class c. However, CAM can only be applied to networks with GAP, and its interpretability is limited by the constraint of the last layer of convolutions.

To address the above problems, Grad-CAM [32] was proposed. The method defined the weight w_k^c of the feature map A^k with respect to the class c as

$$w_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \tag{3}$$

where Z is the number of pixels in activation maps. Due to the computation of partial derivatives concerning the class for obtaining feature importance weights, Grad-CAM can work with neural networks of any structure and depth and is no longer limited by the GAP layer. Grad-CAM generates visualization results for a clearer representation by fusing element-by-element with guided backpropagation dot product [41]. Nonetheless, MIDLN not only adaptively selects significant weights to generate visually better CAMs based on whether the network is more profound or shallower but also feeds back to the network through class activation maps, which in turn changes the network learning strategy and improves the accuracy of the network.

III. Proposed Method

The proposed MIDLN framework is shown in Fig. 3. First, two new SAR oil spill physical features are extracted using the scribble-style interactive sample collection method [42] and composed multiple-feature SAR images with the damping ratio. The MFEHead is employed to extract various physical information, ensuring the uniqueness of the physical information. Next, the backbone of MDCNNet is further downsampled. Feature-extracted results are fused using FPNet [43] and then utilized for the idetect head and AGS-CAExplainer, resulting in localization results and explainer outputs. Finally, based on the explainer results, the interpretability decision coefficient is calculated as signal feedback to the network. The learning strategy of the network is adjusted using the EATStrategy to improve the identification accuracy through iterative updating.

A. Extraction of Multiphysical Backscattering Features

1) Oil Spill Physical Backscattering Features Extraction Methods: To establish an interpretable framework for oil spill detection, in-depth research has been conducted on the physical information embedded in SAR oil spill images. Set σ^0 to denote the image vector of backward scattering coefficients in the SAR image, and the mathematical expression for the backscatter coefficient in SAR images of maritime scenes is given in the following equation:

$$\sigma^0 = \left(\frac{4\pi^2}{\zeta^2}\right) \cdot |K^2| \cdot |\cos(\theta)| \tag{4}$$

where K is the wavevector (also referred to as the scattering vector), ζ is the radar wavelength (usually measured in meters), and θ is the angle of incidence, which is the

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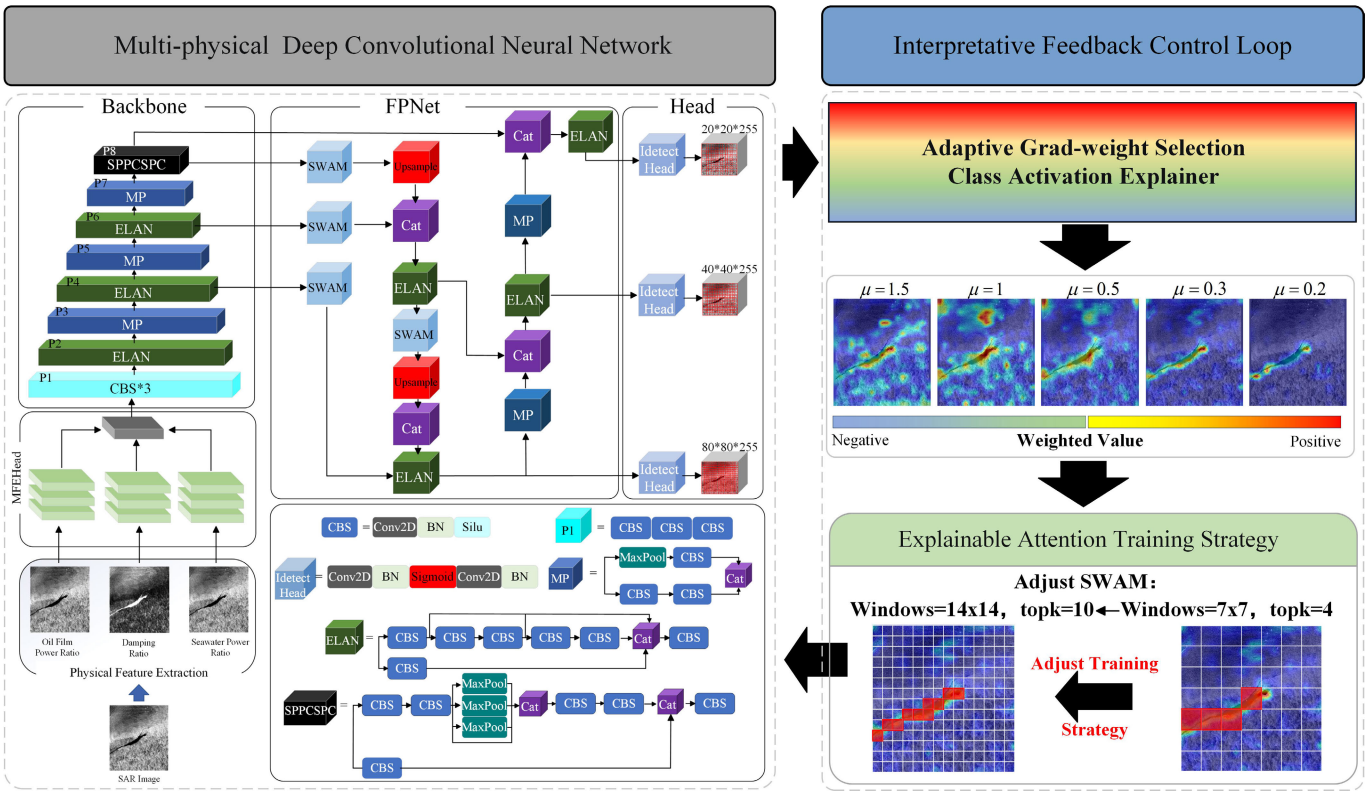


Fig. 3. Framework of MIDLN. The framework is divided into four parts: multiphysics information extraction and input; MDCNNet consisting of Backbone, FPNet, and Idetect Head; AGS-CAExplainer that outputs interpretable results; and the EATStrategy that is used to provide feedback to the network to improve its accuracy.

angle between the radar wave and the surface of an object or terrain. A series of physical features based on the backscatter coefficient have been proposed in the literature [11], [44]. The extraction of single features based on backscattering coefficients, such as oil spill power to mean ratio and background power to mean ratio, requires precise separation and extraction of oil and seawater regions, which makes it challenging to obtain the physical information of SAR oil spill images in complex scenes. Two new oil spill physical features of the oil film power ratio and seawater power ratio of SAR oil spill images are defined for this. The formulas are as follows:

$$p_{\text{oil}} = \frac{\phi_{\text{SAR}}}{\eta_{\text{oil}}} \tag{5}$$

$$p_{\text{water}} = \frac{\phi_{\text{SAR}}}{\eta_{\text{water}}} \tag{6}$$

where p_{oil} is the oil film power ratio and p_{water} is the seawater power ratio. Then, ϕ_{SAR}, η_{oil}, and η_{water} are defined as

$$\phi_{\text{SAR}} = \sqrt{\left(\frac{\sum_{i,j=1}^{n} \sigma_{\text{SAR}}}{n}\right)^2 - \text{mean}(\sigma_{\text{SAR}})^2} \tag{7}$$

$$\eta_{\text{oil}} = \text{mean}\left(\sigma^0_{\text{oil}}\right) \tag{8}$$

$$\eta_{\text{water}} = \text{mean}\left(\sigma^0_{\text{water}}\right) \tag{9}$$

where mean$(\cdot)$ represents the mean of the backscatter power in the respective region, σ_{SAR} denotes the backscatter coefficient of each pixel in the original SAR image, σ^0_{oil} represents the backscatter coefficient of the oil, and σ^0_{water} represents the backscatter coefficient of seawater. Meanwhile, for σ^0 sample



(a) (b) (c)

Fig. 4. Image annotated with per-pixel masks (middle) and sparse scribbles (right). Scribbles are more user-friendly for annotators. (a) SAR oil spill images. (b) Mask annotations. (c) Scribbling interactive sample collection.

acquisition, this article employs the scribble-style interactive sample collection method to obtain representative σ^0 values for oil films and seawater in SAR images [42]. Five randomly sized sample regions are extracted near the center of the object without the need for meticulous delineation of the object's boundaries, which ensures the simplicity and accuracy of the feature extraction process. Fig. 4 illustrates an example of a scribble-style interactive sample collection. The damping ratio is calculated as the ratio of the calm sea surface's backscatter coefficient to the oil film's backscatter coefficient. It can be computed as follows:

$$x_{\text{dr}} = \frac{\sigma^0_{\text{water}}}{\sigma^0_{\text{oil}}} \tag{10}$$

where x_{dr} is the extracted damping ratio feature.

2) Multiphysical Feature Extraction Head: How to fully utilize features is important for identifying oil spills in SAR images. Traditional deep learning methods commonly use shared convolutional weights to extract various oil spill

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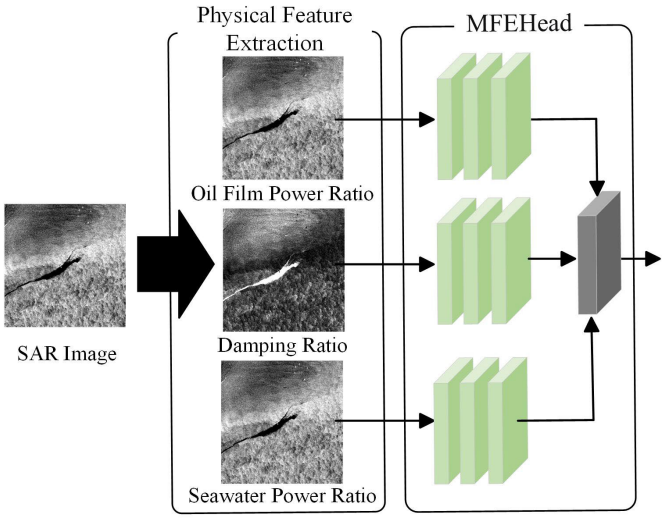


Fig. 5. Designed MFEHead for many different physical features.

features, resulting in nonspecific extracted features, and the shared feature extraction may lead to redundant interference. To address this problem, MFEHead is designed to enhance the uniqueness of extracted physical features and address the effect of interference generated by multiple features sharing convolutional weights on identification results. The framework of MFEHead is shown in Fig. 5.

The traditional convolutional layers aim to learn features in multidimensional space with two spatial dimensions (width and height) and multiple channel dimensions. Therefore, the task of sharing convolutional in the traditional sense is to map cross-channel correlations and spatial correlations simultaneously [45]. The idea behind the multiphysical feature extraction header is to first map the input multiphysical feature data into multiple separate branching spaces that are smaller than the conventional input space. Subsequently, conventional 3×3 convolutions are employed within these individual smaller branch spaces to capture all the correlations and specifics of multiple physical features. A set of 1×1 convolutions then performs the feature integration operation to view the correlation across independent branch channels. Finally, the integrated features are fed into the backbone for further feature extraction.

B. Adaptive Grad-Weight Selection Class Activation Explainer

Considering the correlation between the shape of the oil spill and the background seawater, deep learning can use the background information to assist in model decision-making. In addition, gradient propagation in deep learning may introduce noise. These reasons result in a significant amount of noise in the visual performance of the explainer's results. To alleviate the impact of noise on the explainer's results and obtain higher quality explanations, AGS-CAExplainer is constructed. First, compute the linear prediction score y^c for class c with respect to the feature map A^k of the convolutional layer, i.e., $(\partial y^c / \partial A^k_{ij})$. These gradients flowing back are GAP to obtain feature map importance weights w^c_k in (3). The obtained weights are input into the adaptive weight selection evaluation module. The top m weights are adaptively selected for networks of different depths.

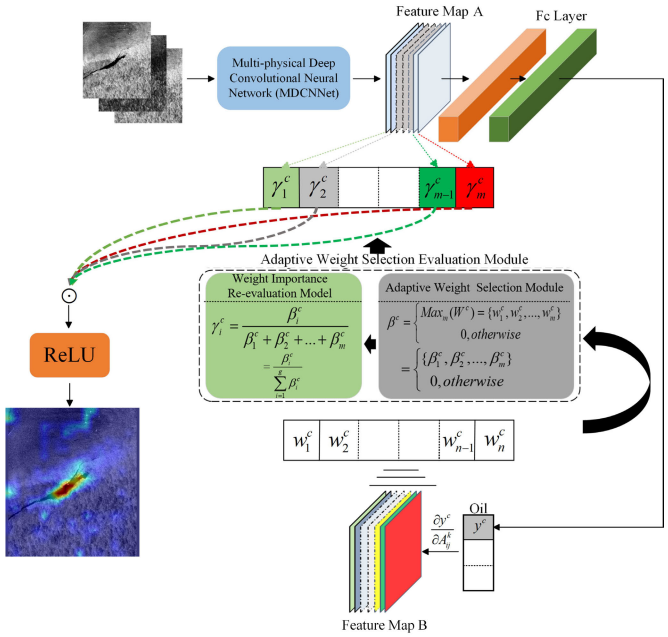


Fig. 6. AGS-CAExplainer overview. The linear prediction score y^c is computed by forward propagating the obtained feature map A. The backpropagation of y^c continues to get a new feature map B, which represents the degree of contribution of each pixel to the class. Feature map B is then subjected to GAP in length and width dimensions to obtain the corresponding weights for feature map A. These weights are input into the adaptive weight selection evaluation module for decision weight selection and importance reevaluation. Subsequently, the newly generated weights are linearly summed with the feature map A and combined with the original image to obtain high-resolution and concept-specific explainer CAMs.

The formula for the adaptive weight selection module is as follows:

$$W^c = \left\{ w^c_1, w^c_2, w^c_3, \ldots, w^c_k \right\} \in \mathbb{R}^{l \times h} \tag{11}$$

$$\beta^c = \begin{cases} b_i, & b_i \in \mathrm{Max}_m\left(W^c\right) \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

where $\mathrm{Max}_m(\cdot)$ selects the top m largest elements in the set, and β^c is the set of the top m important decision weights for adaptive selection, keeping the original position unchanged, and the rest of weights revert to 0. l and h are the width and height of the feature map, respectively. For the top m decision weights, to balance magnitudes of weights and prevent outliers and extreme values from affecting results, input them into the weight importance reevaluation model. The formula for the weight importance reevaluation model is as follows:

$$\gamma^c_i = \frac{\beta^c_i}{\beta^c_1 + \beta^c_2 + \beta^c_3 + \ldots + \beta^c_m} = \frac{\beta^c_i}{\sum_{i=1}^{m} \beta^c_i} \tag{13}$$

where $i \in \{1, 2, 3, .., m\}$ and γ^c_i are new weights generated by the adaptive weight selection module. When the explainer is combined with the target recognition algorithm, the adaptive weight selection evaluation module adaptively selects and evaluates important decision weights, effectively reducing interference from gradient generation or background noise on

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the results and obtaining higher quality CAMs

$$L^c_{\text{AGS-CAExplainer}} = \text{ReLU}\left(\sum_k \gamma_i^c A^k\right). \tag{14}$$

For each image, the corresponding interpretation map E^c is generated by multiplying the $L^c_{\text{AGS-CAExplainer}}$ upsampling with the original image point by point

$$E^c = L^c_{\text{AGS-CAExplainer}} \circ I \tag{15}$$

where $\circ$ refers to the Hadamard product and I is the input image. Finally, the built visual explanation method AGS-CAExplainer is described in Fig. 6. Algorithm 1 describes the complete details of the implementation.

Algorithm 1 AGS-CAExplainer Algorithm

Input: SAR oil spill images X, MDCNNet=$f(X)$, class c, layer b
Output: $L^c_{AGS-CAExplainer}$
initialization;
// get activation of layer b;
// $A_b \leftarrow f_b(X)$;
$C \leftarrow$ the number of channels in A_b;
for k *in* $[0, 1, \ldots, C-1]$ **do**
// $f^c(\cdot)$ as the logit of class c;
$y^c = f^c(X)$ (1)

$w_k^c = \frac{1}{Z}\sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$ (3)

$W^c = \{w_1^c, w_2^c, w_3^c, \ldots, w_k^c\} \in \mathbb{R}^{l\times h}$ (11)

//$Max_m(\cdot)$ selects the top m largest elements in the set;

$$\beta^c = \begin{cases} b_i, & b_i \in Max_m(W^c). \\ 0, & otherwise. \end{cases} \tag{12}$$

$\gamma_i^c = \frac{\beta_i^c}{\sum_{i=1}^m \beta_i^c}$ (13)
end for
$L^c_{AGS-CAExplainer} = ReLU(\sum_k \gamma_i^c A^k)$ (14)

C. Explainable Attention Training Strategy

Two steps are involved in the interpretable attention training strategy, as illustrated in Fig. 7. First, calculate the decision coefficient of AGS-CAExplainer results. When the interpretability decision coefficient reaches a threshold, provide feedback signals to the network, adjust the learning strategy accordingly, and continue iterating updates to enhance the model's accuracy.

1) Interpretability Decision Metrics Calculation: In the early stages of training, the explainer's results display extensive important decision regions (highlighted in red), indicating the network's inclination to learn oil spill global semantic information during the initial training. As training progresses, the important decision region gradually converges to the vicinity of oil spill targets. At this point, the network's ability to learn global feature semantic information reaches its maximum. The decision coefficient is utilized as a feedback signal to change the learning strategy and start the supplementary learning of the shape detail information of the oil spill edge. The decision coefficient of the explainer result is calculated as the timing of the network learning strategy transition, generating the CAM for the oil spill class according to (15). Subsequently, the significance plot for the oil spill class is normalized by converting all intensities above 0 to 1. The explainer result decision coefficient μ is the following formula:

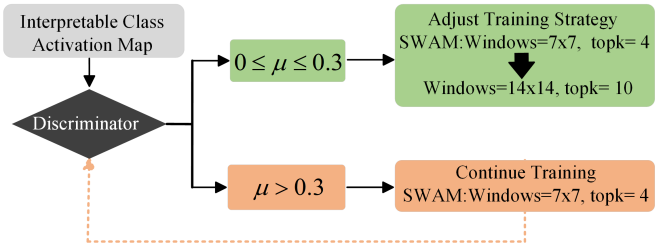


Fig. 7. Explainable attention training strategy flowchart.

$$\mu = |\frac{\text{Area(all)}}{\text{Area(box)}} - 1| \tag{16}$$

where Area(all) denotes the number of nonzero pixels in CAMs and Area(box) is the number of active pixels in the identification ground boxes. When μ approaches 0.3, the network's oil spill global feature semantic information learning capability is close to the maximum. Adjust the network's learning strategy to acquire and supplement detailed edge shape information on the oil spill targets. Otherwise, the original training strategy is maintained, and no processing is performed. The method of tuning learning strategies in decision metric computation is detailed in Section III-C2.

2) Sparse Window Attention Model: The inspiration for the introduced SWAM in this article is derived from the attention mechanism used in Biformer [46], Bi-Level Routing Attention. The attention mechanism maps a query and a set of key–value pairs to an output. SWAM switches between oil spill global feature semantic information learning and oil spill edge shape detail information learning by switching different size windows to change the training strategy. Specifically, when the window is larger, the network filters out the least relevant window key–value pairs by coarse graining the window. It roughly focuses the network's attention region around the oil spill area for learning global feature semantic information. After adjusting the learning strategy, a smaller window is employed for the network to engage in supplementary learning of detailed edge shape information within the oil spill region. In the joint process, the top-k [47] operator is applied to connect windows with high similarity within the window region. Different numbers of attention windows are selected for different window sizes to achieve the learning of oil spill targets in the training phase from coarse-grained to fine-grained asymptotically. The structure of the whole model is shown in Fig. 8.

Given a 2-D input feature map $X \in \mathbb{R}^{H\times D\times C}$, it is first divided into $S \times S$ nonoverlapping windows such that each window O_i' contains the feature mapping of (HD/S^2). The query key–value tensor Q, K, and V for each window is then derived with linear projections

$$Q = O_i' R^q, K = O_i' R^k, V = O_i' R^v \tag{17}$$

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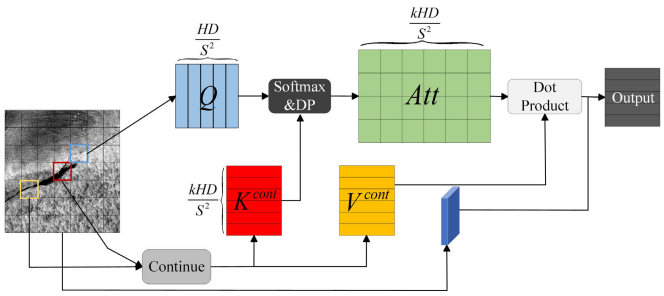


Fig. 8. Details of the SWAM.

where R^q, R^k, and R^v are projection weights of query, key, and value, respectively. Then, the average value of each window is computed for Q and K, respectively, to derive the window's query and key values Q^r and K^r. Matrix multiplication of the transpositions of Q^r and K^r is to derive an adjacency matrix Att of similarity between windows. The adjacency matrix Att measures the degree of semantic similarity between two windows, and the routing index matrix G_r is computed by the topk operator

$$G_r = \text{topk}(\text{Att}). \tag{18}$$

The ith row of G_r contains the first k indices of the most relevant region of the ith window. Continue to collect the key–value tensor

$$K^{\text{cont}} = \text{Continue}(K, G_r),\ V^{\text{cont}} = \text{Continue}(V, G_r). \tag{19}$$

Finally, focus on the collected key–value pairs

$$B' = \text{Attention}\left(Q, K^{\text{cont}}, V^{\text{cont}}\right). \tag{20}$$

Since the transformer attention mechanism lacks local feature learning, a 3×3 convolution is introduced within SWAM

$$B = \text{Attention}\left(Q, K^{\text{cont}}, V^{\text{cont}}\right) + \text{Conv}_{3\times3}(X). \tag{21}$$

The explainable attention training strategy is formally described in Algorithm 2.

Algorithm 2 Explainable Attention Training Strategy

Input: Interpretable class activation maps X_i
Output: Training Strategy
initialization;
for epochs **do**;
  if $\mu > 0.3$ **then**
    //Continue with the original training strategy.
    Windows=7, Topk=4
  if $0 \le \mu \le 0.3$ **then**
    //Adjust Training Strategy.
    Windows=14, Topk=10 ← Windows=7, Topk=4
end for

D. Loss Function

In SAR oil spill images, oil spill targets tend to concentrate around the center of the bounding box. In addition, oil spill areas often exhibit elongated and punctate shapes. Therefore, this article improves a weighted localization loss based on the YOLOv7 localization loss to achieve more accurate oil spill identification results

$$L_{\text{locge}} = (1-\lambda)L_{\text{loc}} + \lambda L_{\text{ge}} \tag{22}$$

$$L_{\text{loc}} = 1 - \text{IOU} + \frac{\rho^2\left(b, b^{\text{gt}}\right)}{d^2} + \alpha v \tag{23}$$

where $\alpha = (v/(1-\text{IOU})+v)$, $\text{IOU} = 1-(l_{\text{box}} \cap l'_{\text{box}}/l_{\text{box}} \cup l'_{\text{box}})$, l_{box} are predicted localization boxes, and l'_{box} are ground-truth boxes. $v = (4/(\pi^2)(\arctan(w^{\text{gt}}/h^{\text{gt}}) - \arctan(w/h))^2$. b and b^{gt} represent the centers of predicted localization boxes and ground-truth boxes. ρ represents the computation of the Euclidean distance between two center points. d denotes the diagonal distance of the smallest outer rectangular region containing the predicted positioning boxes and the ground-truth boxes

$$L_{\text{ge}} = \exp\left(-\frac{\sqrt{W_2^2(N_a, N_b)}}{O}\right) \tag{24}$$

$$W_2^2(N_a, N_b) = \left\| \left[c_{xa}, c_{ya}, \frac{w_a}{2}, \frac{h_a}{2}\right]^T, \left[c_{xb}, c_{yb}, \frac{w_b}{2}, \frac{h_b}{2}\right] \right\|_F^2 \tag{25}$$

where L_{loc} represents the original localization loss in YOLOv7 and λ is the weight parameter. c_{xa}, c_{ya}, $(w_a/2)$, and $(h_a/2)$ are the coordinates for centers of the predicted localization boxes and their length and width, respectively, and c_{xb}, c_{yb}, $(w_b/2)$, and $(h_b/2)$ are the coordinates for centers of ground-truth localization boxes and their length and width, respectively. $\|\cdot\|_F$ is the Frobenius norm. O is a constant closely related to the dataset. $W_2^2(N_a, N_b)$ is a distance metric and cannot be used directly as a similarity metric. Therefore, a new localization metric L_{ge} has been obtained utilizing its exponential form. Confidence loss and classification loss formulas are as follows:

$$L_{\text{loss}} = -\frac{1}{N}\sum_{i=1}^{N}(y_i \log(p_i) + (1-y_i)\log(1-p_i)) \tag{26}$$

where L_{loss} is L_{cls} and L_{conf}, p_i is the network prediction, y_i is the ground truth, and N is the number of classes. The total loss function formula is shown as follows:

$$\text{Loss}_{\text{oil}} = L_{\text{locge}} + L_{\text{cls}} + L_{\text{conf}} \tag{27}$$

where Loss_{oil} is the total loss, L_{locge} is the improved weighted localization loss, L_{cls} is the loss of classification, and L_{conf} is the loss of confidence. Finally, oil spill accurate localization boxes are generated by minimizing the loss function during training.

IV. Experiments

In this section, a series of experiments are conducted to validate the performance of MIDLN and demonstrate its interpretative effectiveness. First, a comparison experiment of oil–water features in the SAR oil spill is designed, and the validity of two new oil spill physical features and the comparison experiment is verified by synchronously using the network. Next, MIDLN is compared with several state-of-the-art detection frameworks and experimental evaluations are conducted. Then, the interpretability accuracy of various

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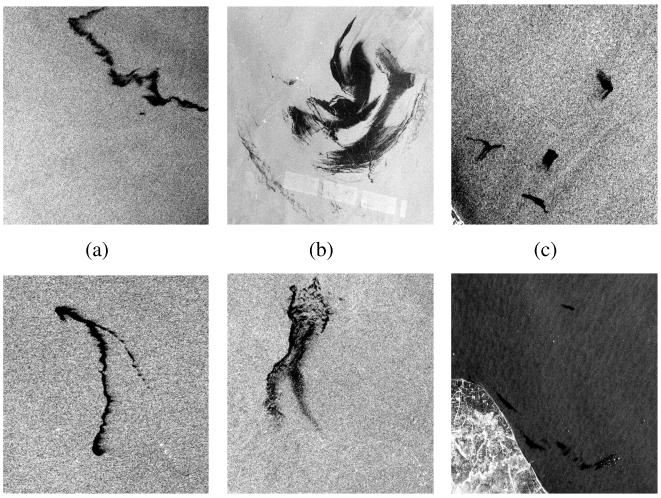


Fig. 9. Examples of oil spill datasets under various imaging states. (a) and (d) Long strip of oil spills. (b) and (e) Oil spills without clear boundaries against a bright background, exhibiting highly irregular shapes. (c) Oil spills appear in many small scattered spots. (f) Oil spills on a dark background.

explainers in localization is tested. Finally, the effectiveness of MFEHead, SWAM, and EATStrategy is validated, and recognition evaluations from different metrics are shown in the following.

A. Datasets

In the experimental work, Sentinel-1 datasets for 82 scenes during the period 2014–2021 were obtained from the Alaska Satellite Facility (ASF) Data Search website (https://search.asf.alaska.edu) [1], [48]. Sentinel-1 data from the single-look complex (SLC) products with L1 interferometric wide swath (IW) strip scan mode in the vertical–vertical (VV) and vertical–horizontal (VH) polarization modes are utilized in this work to extract various features. The obtained data underwent thermal noise removal, radiometric calibration, filtering, terrain correction, and multiview angle preprocessing, followed by cropping out the oil spill areas. The final SAR oil spill images are obtained at a resolution of 20×20 m with 1024×1024 pixels.

The total dataset consists of 105 SAR oil spill images, with 70% of the training set and 30% of the test set. There are 73 training sets and 32 test sets. Some sample images are shown in Fig. 9. The Make Sense website is utilized to mark up ground-truth labels.

B. Experimental Setup

All experiments are conducted under Windows 11 using Python 3.8, PyTorch 1.11.1, and CUDA 11.6 and run on a GeForce RTX 4060 GPU. The standard stochastic gradient descent trains the network with a momentum of 0.937, a weight decay of 0.0005 for the Nesterov accelerated gradients utilized, and a batch size of 4. The learning rate is initially set to 0.01, and the entire training process involves 150 epochs.

C. Evaluation Criteria

To objectively evaluate the performance of different models, various metrics are employed to compare the identification results. The recall, precision, $F1$, and mean average precision (mAP) serve as object-level accuracy metrics for evaluating the performance of the methods under comparison. The calculation of precision and recall metrics is defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{28}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{29}$$

where true positives (TP) and true negatives (TN) indicate correct predictions and false positives (FP) and false negatives (FN) indicate incorrect results. Higher precision means higher detection accuracy. The higher the recall, the more positive samples are detected. The $F1$ is a balanced reflection of the accuracy of the algorithm by being able to consider both precision and recall values

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{30}$$

The mAP is a composite metric obtained by averaging the AP values, which uses an integral method to calculate the area enclosed by the precision–recall curve and coordinate axis of all categories. Hence, the mAP equation is shown as follows:

$$\text{mAP} = \frac{\text{AP}}{N} = \frac{\int_0^1 P(R) dR}{N} \tag{31}$$

where P denotes the precision, R denotes the recall, and N is the number of classes.

D. Comparative Analysis and Selection of Oil–Water Features

As input to the interpretable identification network, features discussed in this article are pixel-based. Referring to [11], [28], and [44], eight features are chosen to compare with our proposed features in this article, including five of which are based on the backscattering coefficients, namely, oil spill mean, oil spill standard deviation, backgrounds mean, backgrounds standard deviation, and damping ratio. Three polarimetric features are based on polarimetric scattering angles, namely, H, A, and $Alpha$. Meanwhile, to align with this article's AFValue oil–water feature comparative analysis method, the features based on backscattering coefficients combine to generate two kinds of SAR oil spill features, with mean and standard deviation as the two new backscattering feature naming forms, which are discussed in comparison with the oil film power ratio and seawater power ratio. Features extracted from SAR oil spill images are normalized to the same dimension. The candidate features are analyzed to select the most appropriate feature subset.

1) Seven SAR oil spill images are selected for each of eight features, and samples of the oil spill and seawater areas are randomly selected using the scribble-style sample collection method.
2) Average eigenvalues of all pixels of samples in each feature are calculated. Specifically, two average eigenvalues, the average eigenvalue of oil spill and the average eigenvalue of seawater, will be obtained for each of

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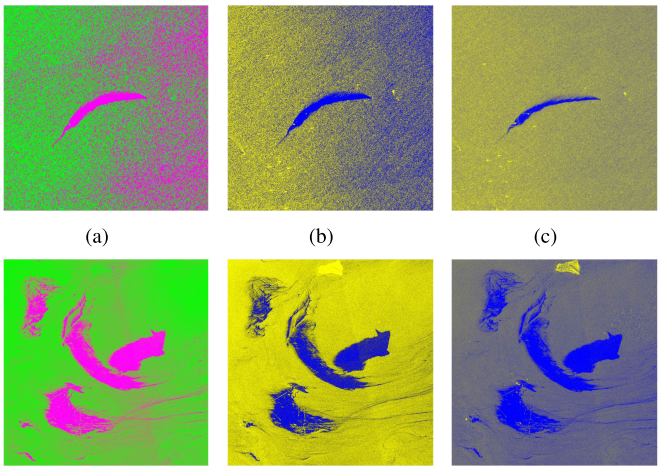


Fig. 10. Different feature subset visualization results. (a) and (d) Visualization results for *H, A*, and *Alpha* of **Subset 1**. (b) and (e) Visualization results for mean, standard deviation, and damping ratio of **Subset 2**. (c) and (f) Visualization results for oil film power ratio, seawater power ratio, and damping ratio of **Subset 3**.

seven images for each feature, for a total of 14 average eigenvalues.

3) To judge the advantages and disadvantages of different features, this article designs a feature mean normalization index. Evaluate the degree of difference in eigenvalues on oil film and seawater in different features. Analyze and compare eight features to identify the optimal subset of features

$$\text{AFValue} = \frac{|\sigma_{\text{sea}} - \sigma_{\text{oil}}|}{\sigma_{\text{sea}}} \tag{32}$$

where σ_{oil} and σ_{sea} are the AFValue of the oil spill and the AFValue of seawater in SAR images after feature extraction, respectively. Evaluate the quality of features by comparing the difference in AFValue between the oil film and seawater across various features.

The obtained visualization combinations of different oil spill physical features are shown in Fig. 10. Where *H/A/Alpha* constitute Fig. 10 (a) (d) **Subset 1**, mean, standard deviation, and damping ratio constitute Fig. 10 (b) (e) **Subset 2**, and oil film power ratio, seawater power ratio, and damping ratio constitute Fig. 10 (c) (f) **Subset 3**. The $AFValues$ oil-water feature comparative analysis experiments are shown in Fig. 11. Among all the features extracted, the $AFValues$ of the seawater power ratio is higher, indicating that the feature values of oil film and seawater in this feature are highly different. Thus, the oil spill area can be identified more accurately. The $AFValues$ of the oil film power ratio and the damping ratio are pretty close but slightly lower than the seawater power ratio. The $AFValue$ of these three oil spill features mentioned above are significantly higher than those of the original image, allowing for a good distinction between oil spill areas and the water surface. The $AFValue$ of the mean is lower than that of the original image, indicating a weaker differentiation between the oil film and seawater under this feature. The $AFValue$ of the standard deviation is higher than that of the original image. However, it is weaker than the oil film power ratio, damping ratio, and seawater power ratio. The $AFValues$ based on polarized scattering angles *H/A/Alpha* fluctuate around the original image, and all these features are smaller than the oil film power ratio, damping ratio, and seawater power ratio. To validate the effectiveness of two new oil spill physical features and assess the feasibility of $AFValues$ oil-water feature comparative analysis experiments, a combination experiment with the proposed MFEHead is carried out.

TABLE I

QUANTITATIVE COMPARISON OF INPUT NETWORKS WITH DIFFERENT FEATURE SUBSETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Oil Spill Features	$mAP_{50}(\%)$	$mAP_{50:95}(\%)$
Original images	86.69	**53.97**
Subset 1	86.93	52.14
Subset 2	87.67	52.18
Subset 3	**89.31**	53.48

Inputting the above subsets into the network, as shown in Table I, **Subset 3** achieves the highest accuracy, with mAP_{50} reaching 89.31%. This result demonstrates the effectiveness of the oil film power ratio and seawater power ratio. Simultaneously, the oil–water feature comparative analysis method aligns with the accuracy of deep learning results, confirming the feasibility of the feature comparative analysis method. Based on the backscatter coefficient, the oil–water feature comparative analysis method fundamentally illustrates the differences between various features, enhancing the model's interpretability. The seawater power ratio and oil film power ratio weaken the background noise in SAR images, enrich the feature information of oil spill targets, and effectively maintain the spatial consistency and feature diversity of the oil spill area. Deep learning based on the image pixel level also pays more attention to the difference in backscattering coefficient intensity between oil and water in SAR oil spill images. The analysis selected the oil film power ratio, seawater power ratio, and damping ratio as the best feature set.

E. Comparison With Other Typical Models

This article compares classical supervised recognition methods, including Faster R-CNN [19], YOLOv4 [24], YOLOX [26], YOLOv5 [25], Mask R-CNN [49], YOLACT [50], YOLOV7 [29], and MFSCNet [28]. The oil spill identification results are shown in Fig. 12, and the assessment results are listed in Table II (with the best results shown in bold). As shown in Fig. 12, the most challenging aspect of oil spill detection lies in the oil spill's irregular shape and weak boundaries. MIDLN exhibits a significant advantage in recognizing long strip-shaped oil spills compared to other methods. Similarly, MIDLN also performs well in the misjudgment and misjudgment of some oil spill areas with unclear boundaries and small oil spill areas that are difficult to identify. The non-interpretability of traditional deep learning cannot generate an interpretation of the target to provide feedback to adjust the model's learning strategy in the face of a non-regular state of oil spill targets, resulting in low localization accuracy. Faster R-CNN performs two rounds of resampling through RoIAlign on feature maps before object detection, which can lead to poorer predictions in edge regions

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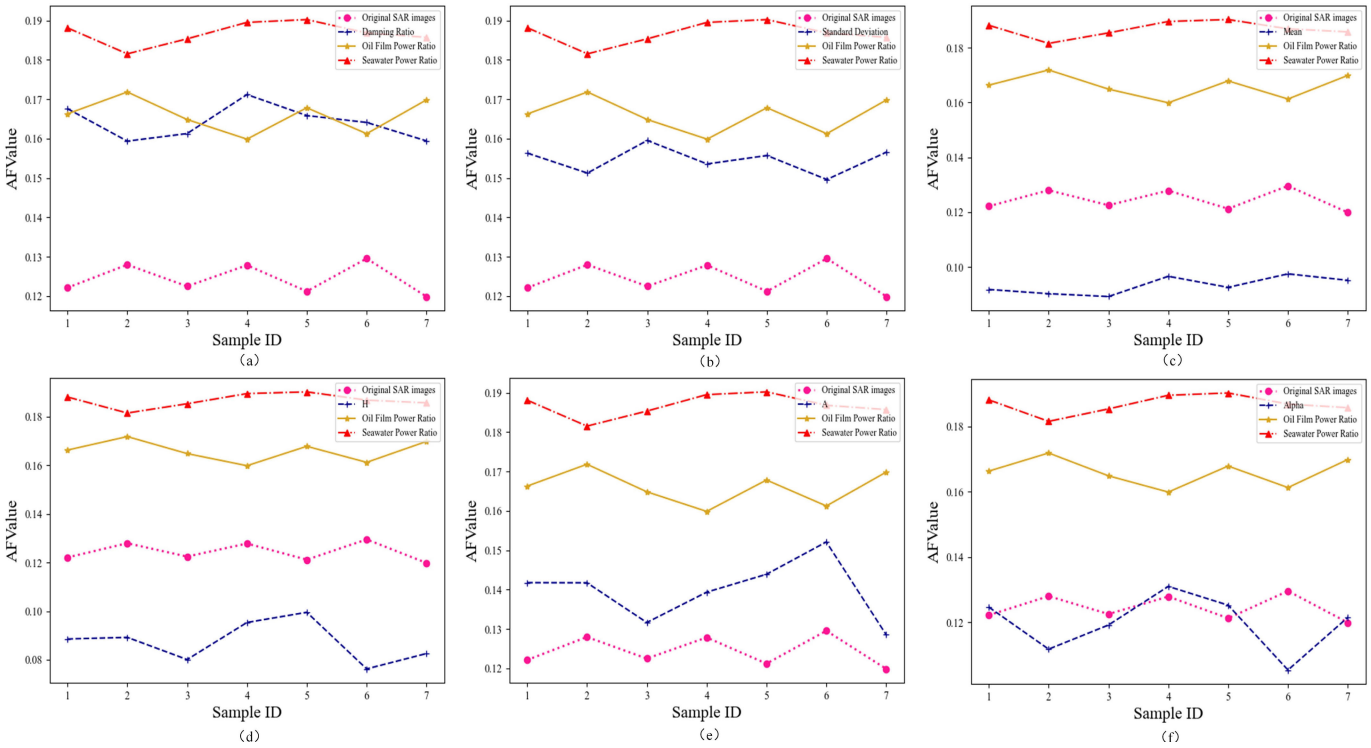


Fig. 11. Quantitative comparison of AFValue of features of the oil spill from multiple SAR images with original images, oil film power ratio, and seawater power ratio (higher is better). (a) Damping ratio. (b) Mean. (c) Standard deviation. (d) *H*. (e) *A*. (f) *Alpha*.

TABLE II
QUANTITATIVE COMPARISON OF MIDLN WITH OTHER OIL SPILL IDENTIFICATION TECHNIQUES ON THE SAME DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	$mAP_{50}(\%)$	$mAP_{50:95}(\%)$	$F1(\%)$	$Precision(\%)$	$Recall(\%)$	$Times$(s)	$GFLOPs$
Fast R-CNN [19]	54.76	-	-	-	-	0.17	168.41
YOLOv4 [24]	61.47	-	-	-	-	0.31	360.64
Mask R-CNN [49]	55.58	-	57.29	-	-	0.24	206.10
YOLOX [26]	79.05	-	-	-	-	0.30	66.98
YOLACT [50]	73.88	-	72.63	-	-	0.13	338.17
YOLOv5 [25]	84.29	52.81	80.16	85.91	75.43	0.141	**48.2**
YOLOv7 [29]	86.60	54.91	82.11	85.49	77.58	**0.0629**	105.1
MFSCNet [28]	86.24	-	79.10	-	-	0.31	141.60
MIDLN	**94.00**	**59.92**	**88.12**	**94.71**	**86.63**	0.0804	130.4

and is prone to the problem of misclassification. YOLOv5 reduces the image's resolution due to the presence of the focus module, removing spatial spacing information for small targets in SAR images and thereby increasing the detection error for missing targets. Although YOLOv7 obtains rich semantic information through the ELAN [51] module, it is still susceptible to background noise when recognizing fuzzy boundaries and long strip-shaped oil spills. MFSCNet utilizes multifeature SAR images to alleviate background noise, but recognition results are affected by redundant interference in shared convolutional feature extraction.

MIDLN considers the physical properties of SAR oil spill images, and the proposed two new oil spill physical features effectively enhance the features and spatial consistency of the oil spill areas, reducing the impact of scattering noise on identification accuracy in SAR images. MFEHead preserves the specificity of oil spill physical information, alleviating redundant interference from multiple features. The addition of the explainer effectively enhances the interpretability of the model. In addition, utilizing explainer results to calculate decision coefficients as feedback signals allows timely and effective integration of explanations into the model's learning process. By switching the network's learning strategy, the identification accuracy and reliability of the model are improved. As shown in Table II, MIDLN's mAP_{50} reaches 94.00% and $mAP_{50:95}$ reaches 59.92%, while the precision, recall, and $F1$ are all better than other oil spill detection algorithms.

F. Harnessing Explanations for Object Localization

This section demonstrates the effectiveness of ASC-CAExplainer for class-conditional localization of the oil spill in a given SAR image. Similar to the calculation of decision coefficients in EATStrategy, the oil spill-specific saliency map L_{oil} is normalized using the explanation map E^c generated in (14). The difference is setting a threshold δ that converts all intensities above δ to 1. The following equation defines the pixel-level cross-linking metrics for the

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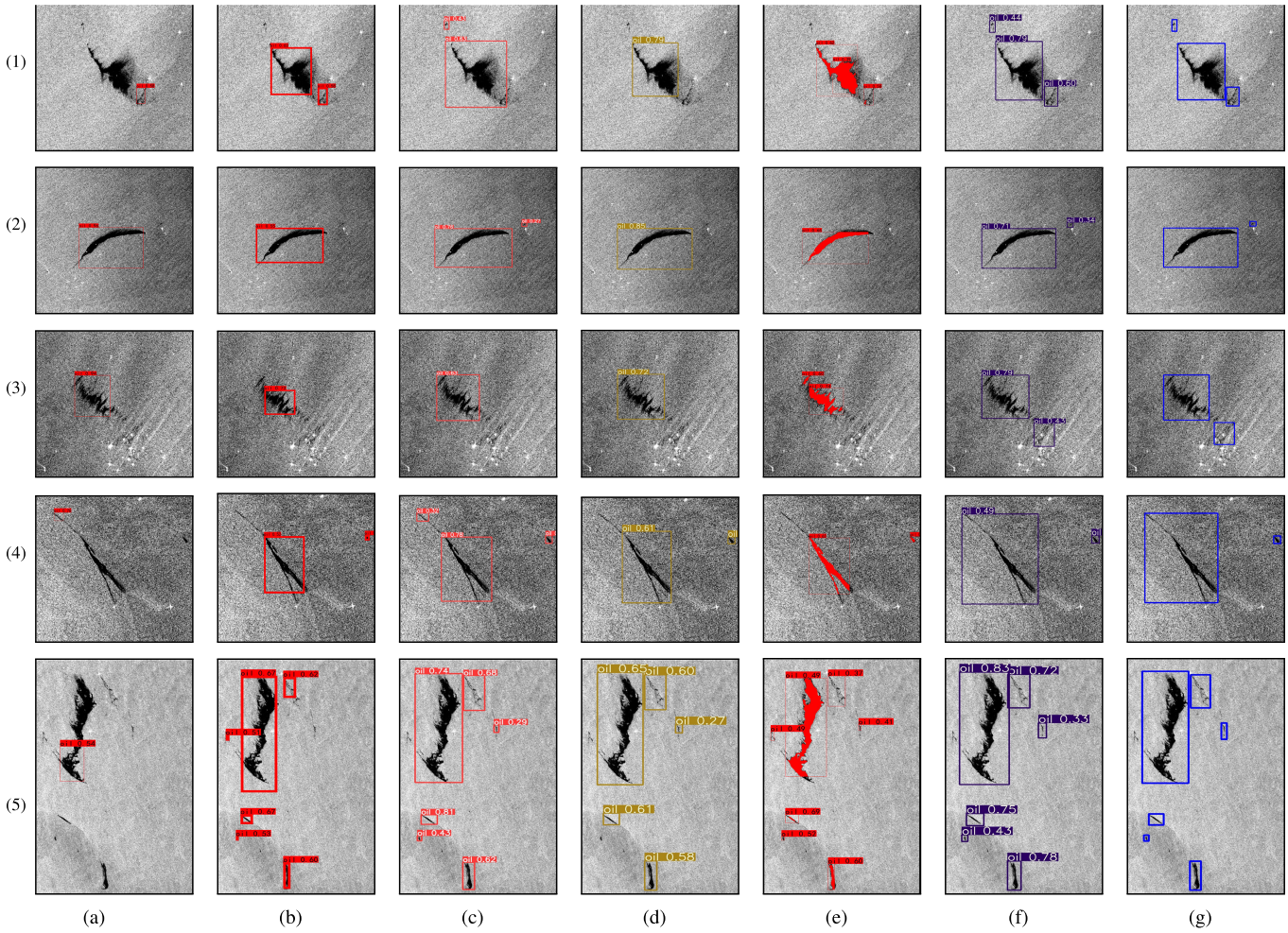


Fig. 12. Visual comparison of oil spill identification on the Sentinel-1 dataset (MFSCNet is the localization segmentation model, and the rest of the comparison models are localization models). (a) Faster R-CNN. (b) YOLOX. (c) YOLOv5. (d) YOLOv7. (e) MFSCNet (simultaneous localization and segmentation). (f) MIDLN. (g) Oil spill original images and ground-truth labels.

oil spill category, threshold δ, and image I:

$$\mathrm{Loc}_I^{\mathrm{oil}}(\delta) = \frac{\mathrm{Area(internal\ pixels)}}{\mathrm{Area(boundingbox)} + \mathrm{Area(externalpixels)}} \tag{33}$$

where Area(bounding box) refers to the area of the oil spill ground-truth boxes in the given image I, Area(internal pixels) denotes the number of nonzero pixels of the interpretation map that lie inside the bounding boxes, and Area(external pixels) denotes the number of nonzero pixels that lie outside the bounding boxes. Higher values of $\mathrm{Loc}_I^{\mathrm{oil}}(\delta)$ indicate better explainer result localization performance. To fairly compare the interpretation performance of different explainers, the same δ is unitized as the threshold value for the results of all four explainers (Grad-CAM [32], XGrad-CAM [34], Grad-CA$M++$ [33], and AGS-CAExplainer). Evaluation results are shown in Table III, AGS-CAExplainer's $\mathrm{Loc}_I^{\mathrm{oil}}(\delta = 0)$ reaches 0.3516 and $\mathrm{Loc}_I^{\mathrm{oil}}(\delta = 0.25)$ reaches 0.3384. As shown in Fig. 13, AGS-CAExplainer outperforms the other explainer methods in all aspects. The activation map has more than 70% of important decision areas (red and yellow-green areas) falling in the oil spill ground-truth boxes. AGS-CAExplainer adaptively selects feature weights important to the oil spill class and balances their interrelationship. It retains reliable target localization information while removing background noise to obtain high-quality interpretation results.

TABLE III

QUANTITATIVE COMPARISON OF AGS-CAEXPLAINER WITH OTHER INTERPRETABLE TECHNIQUES ON IDENTIFICATION (HIGHER IS BETTER). $\mathrm{mLOC}_I^{\mathrm{OIL}}(\delta)$ IS THE AVERAGE $\mathrm{LOC}_I^{\mathrm{OIL}}(\delta)$ PER LABEL PER IMAGE

Methods	$mLoc_I^{oil}(\delta = 0)$	$mLoc_I^{oil}(\delta = 0.25)$	$mLoc_I^{oil}(\delta = 0.5)$
Grad-CAM [32]	0.2168	0.1684	0.0886
xGrad-CAM [34]	0.1176	0.0651	0.0625
Grad-CAM++ [33]	0.1982	0.1886	0.1454
AGS-CAExplainer	**0.3516**	**0.3384**	**0.2954**

G. Exploration of Training Efficiency for Oil Spill Identification

Oil spills, being low-probability marine disasters, present significant challenges in collecting datasets. To explore the minimum data sample requirements for the model and assess whether it is underfitting. The 105 oil spill images are redistributed to form nine sets for data analysis experiments. The training set size is 1%, 2%, 5%, 10%, 20%, 50%, 70%, 80%, and 90% of the total datasets. To illustrate the relationship between the number of training samples and the efficiency of MIDLN, in this article, relevant experiments are carried

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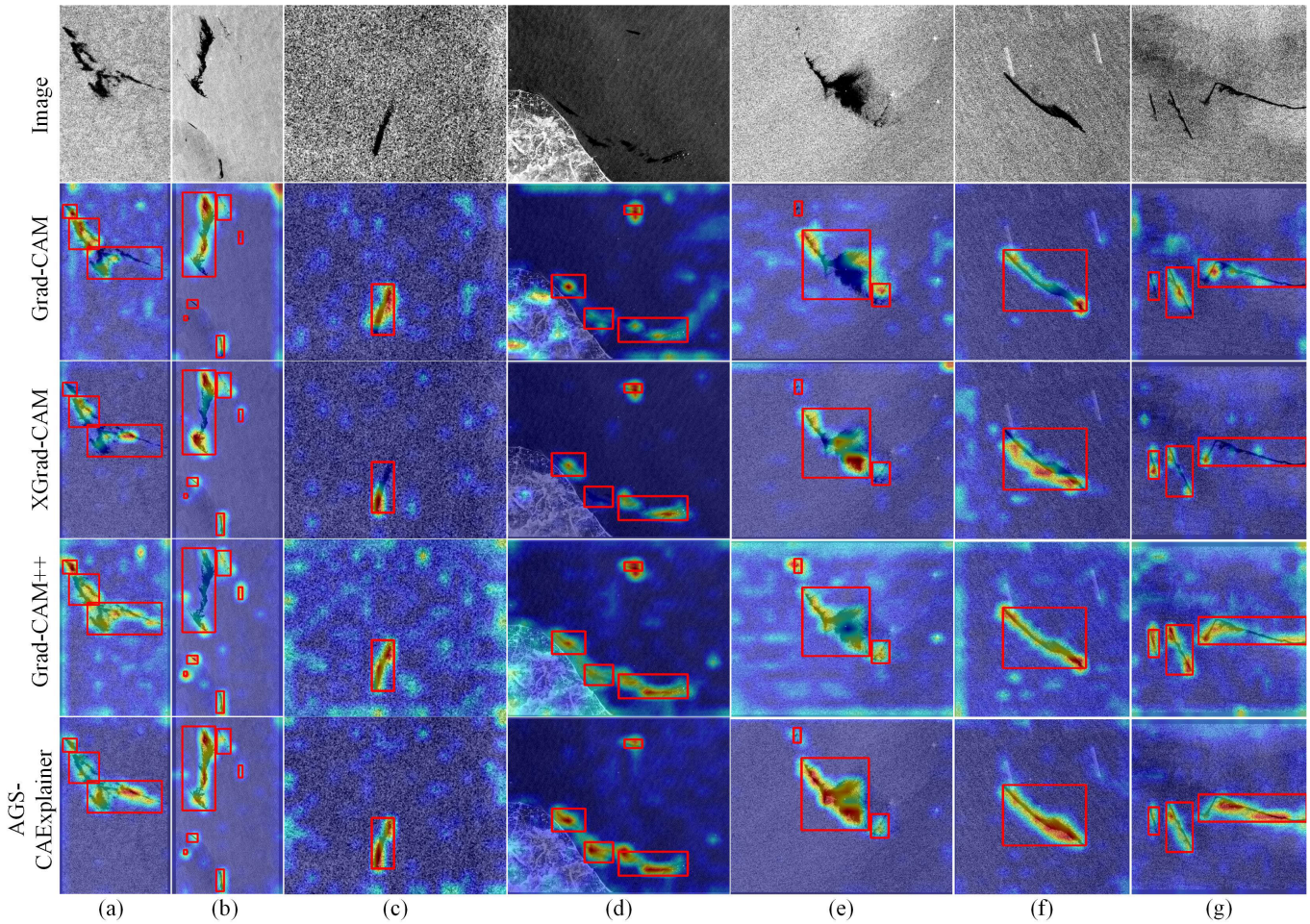


Fig. 13. Object localization capabilities of Grad-CAM, XGrad-CAM, Grad-CAM++ and AGS-CAExplainer. (a) and (b) Different explainer results under multiple oil spill targets. (c) Different explainer results when the background coherent spot noise is large. (d) Different explainer results on a dark background. (e) and (f) Different explainer results under the long strip-shaped oil spill targets. Red boxes represent ground truth annotations for the SAR images.

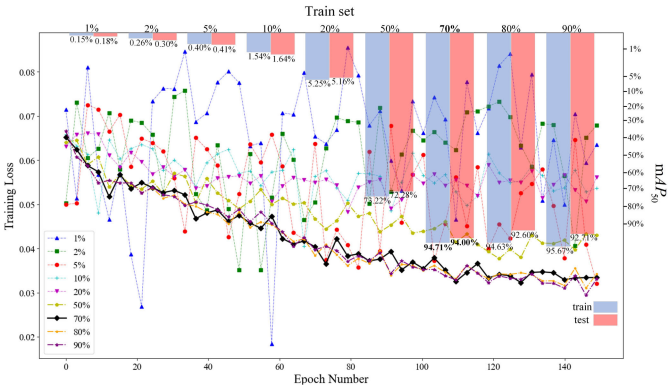


Fig. 14. Training loss curves and train and test mAP_{50} (blue: train and red: test) for different training sets. The settings in this article are shown in bold.

out, and Fig. 14 shows the model's training loss curve under different training sample sizes, along with the corresponding training and test accuracies.

As shown in Fig. 14, during training, the training loss curve for the experimental group with fewer data exhibits a lack of convergence and decline, maintaining fluctuations around the initial training values. This suggests that the model has not learned sufficient features. When the dataset is allocated in a 70% ratio, the training loss curve demonstrates a noticeable descent, followed by gradual stabilization during the training process, fluctuating around 0.033. Simultaneously, there is a slight disparity in the mAP_{50} between the training and test sets, indicating the absence of overfitting. With the continuous increase in the training dataset, it is observed that the descending trend of the training loss function is similar to that when the training set is 70%, and it also stabilizes around 0.033. Therefore, the dataset allocation setting of 70% can yield favorable training results for MIDLN.

H. Ablation Study

1) Comparison of Different Modules: Ablation studies aim to analyze the effects of modules and different hyperparameter loss functions to evaluate the contribution of modules better. YOLOv7 is chosen as the baseline model. The MFEHead, SWAM, and EATStrategy are introduced to the baseline. The effects of the modules proposed in this study on the oil spill detection accuracy are presented in Table IV.

As indicated in the second row of Table IV, the incorporation of physical information and MFEHead effectively mitigates the impact of speckle noise and multifeature redundancy interference on oil spill detection. Compared with the baseline model, mAP_{50} is improved by 2.71%. This suggests that the proposed MFEHead and the two new oil spill physical features significantly enhance oil spill detection accuracy.

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TABLE IV

EFFECT OF THE DESIGN MODULE ON SAR OIL SPILL IMAGES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Baseline	Subset3+MFEHead	SWAM	EATStrategy	$mAP_{50}(\%)$	$mAP_{50:95}(\%)$
✓				86.60	54.91
✓	✓			89.31	53.48
✓		✓		87.49	55.61
✓	✓	✓		92.50	57.40
✓	✓	✓	✓	**94.00**	**59.92**

TABLE V

CHOICE OF THE WEIGHTING FACTOR λ IN THE LOSS FUNCTION. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Weighting factor	$mAP_{50}(\%)$	$mAP_{50:95}$	*F1* (%)	*P* (%)	*R* (%)
$\lambda = 0.0$	89.25	56.50	84.32	89.54	86.36
$\lambda = 0.2$	88.20	56.40	85.22	83.58	90.14
$\lambda = 0.4$	87.79	54.39	81.10	75.94	87.82
$\lambda = 0.6$	**94.00**	**59.92**	**88.12**	**94.71**	**86.63**
$\lambda = 0.8$	83.49	45.80	83.21	81.74	75.49
$\lambda = 1.0$	83.11	44.53	76.00	75.72	76.91

TABLE VI

DIFFERENT THRESHOLD CHOICES FOR EXPLAINABLE ATTENTION TRAINING STRATEGY. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Thresholds	$mAP_{50}(\%)$	$mAP_{50:95}$	*F1* (%)	*P* (%)	*R* (%)
7×7-4 (NO-EATStrategy)	92.50	57.40	86.00	91.21	86.56
14×14-10 (NO-EATStrategy)	92.12	59.41	87.61	90.11	83.14
$\mu = 0.2$ (EATStrategy)	91.60	58.42	86.54	86.14	85.96
$\mu = 0.3$ (EATStrategy)	**94.00**	**59.92**	**88.12**	**94.71**	**86.63**
$\mu = 0.5$ (EATStrategy)	85.31	57.82	81.68	81.25	79.96
$\mu \geq 1$ (EATStrategy)	90.70	57.73	84.12	77.65	90.81

TABLE VII

COMPARISON OF DIFFERENT FEATURE EXTRACTION LAYERS IN MFEHEAD. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Set	$mAP_{50}(\%)$	$mAP_{50:95}$	*F1* (%)	*P* (%)	*R* (%)
NO-MFEHead(150)	87.91	52.91	83.51	85.23	81.54
MFEHead2-3×3	88.62	58.13	85.54	91.32	80.38
MFEHead3-3×3	**94.00**	**59.92**	**88.12**	**94.71**	**86.63**
MFEHead4-3×3	85.51	51.43	82.35	80.06	84.82

EATStrategy acquires feedback signals by computing decision coefficients in explainer results. By switching between learning strategies for oil spill global feature semantic information and oil spill edge shape detail information, it effectively mitigates misclassifications and missing judgments of irregularly shaped oil spills and enhances the interpretability and performance of the model. After using EATStrategy, mAP_{50} and $mAP_{50:95}$ increased by 1.5% and 2.52%, respectively. Finally, MIDLN achieved mAP_{50} and $mAP_{50:95}$ of 94.00% and 59.92%, respectively.

2) Comparison of Different Hyperparameters: In $Loss_{oil}$, L_{locge} is the way of weighting, and different weighting factor values produce different identification results. The identification results with different weighting factors are compared, as shown in Table V. Oil spills show irregular shape and size, and with the increase of the weight factor, the localization accuracy shows a trend of first increasing and then decreasing. The localization accuracy reaches the highest when $\lambda = 0.6$. Therefore, the weight factor of the localization loss function is set to 0.6.

EATStrategy adjusts the model's learning strategy based on the decision coefficient calculated from the explainer results, achieving an interpretable feedback loop and improving network performance. Nevertheless, arbitrarily changing the learning strategy may prove counterproductive during training and learning. If the learning strategy is altered too early, the network may erroneously learn redundant background features. Conversely, changing the strategy too late may result in insufficient fine-grained learning. To explore the optimal timing for learning strategy transitions, experiments investigate the impact of switching learning strategies on recognition results under different decision coefficients μ, as shown in Table VI.

The total number of training rounds is 150. In Table VI, the first and second rows present results obtained by setting window sizes of 7×7 and 14×14, respectively, selecting the top four or ten windows with the highest similarity for attention and training continuously for 150 rounds. The third to sixth rows display results using the interpretable training strategy. With the explainer's decision coefficient set to $\mu = 0.3$, mAP_{50} and $mAP_{50:95}$ improved by 1.5% and 2.52%, respectively, compared to the traditional training method. Therefore, setting $\mu = 0.3$ is identified as the optimal timing for shifting the training strategy.

3) Effect of Different Numbers of Layer Convolutions in MFEHead on Results: The number of convolutional layers in the MFEHead impacts the localization results. Generally, shallow features in deep learning models are more susceptible to the interference of coherent scattering noise in SAR images. Therefore, experiments with different numbers of feature extraction layers are set up to identify the MFEHead that matches the oil spill backscattering features. The identification results of MFEHead with varying numbers of feature extraction layers are compared, as shown in Table VII.

The utility of MFEHead, designed for multiple features in this article, is validated in the first row of Table VII. The network's performance and stability significantly decrease when MFEHead is not applied. In addition, Table VII verifies the effect on the identification results when the number of feature extraction layers is 2 (2: 3×3), 3 (3: 3×3), and 4 (4: 3×3). The highest identification accuracy is observed when the number of feature extraction layers is 3 (3: 3×3), with mAP_{50} reaching 94.00% and $mAP_{50:95}$ reaching 59.92%. Therefore, in this article, it is determined to use MFEHead with feature extraction layers of 3 for the preliminary extraction of physical information from multiphysical SAR oil spill images.

V. CONCLUSION

This article proposes an oil spill identification framework based on interpretable feedback. The framework maximizes the utilization of explainer results to provide feedback signals to the network. By switching network learning strategies, it effectively reduces instances of missed and false detections of oil spill targets, enhancing both the interpretability and recognition accuracy of the network. Meanwhile, to enhance the physical interpretability of the model, a scribble-style sample collection method is employed to simplify the process of extracting physical information related to oil spills. Combined with the multiple physical feature extraction head, it enhances the consistency and uniqueness of oil spill features, reducing the impact of feature redundancy on the results. Experimental results show that MIDLN can better deal with long

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strip-shaped and small oil spill areas with fuzzy boundaries in multitargets. The embedding of explainers and oil spill physical information further enhances the physical interpretability of the network. By utilizing high-quality explainer results to provide feedback to the network, the phenomenon of missing strip-shaped oil spill judgments and misclassification small oil spill targets are effectively addressed. In the future, more appropriate polarimetric features of oil spills based on Sentinel-1 SAR images can be adequately studied. In addition, it is important to consider the interaction between the oil–water interface during rough sea conditions, which can result in the blurring of oil spill edges.

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